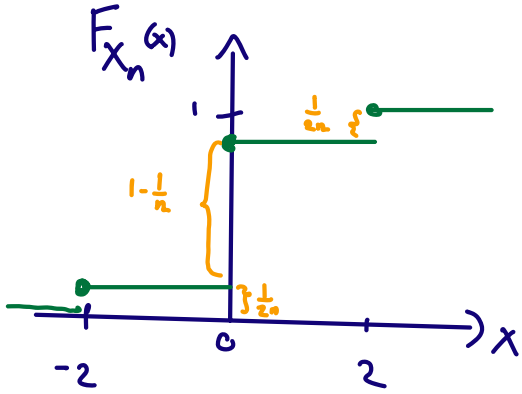
Omtenta SF2940 HT23

1) a.) see book or notes.

b.) The CDF of the random variable is

$$F_{X_n}(x) = \begin{cases} 0 & \text{if } x < -2, \\ \frac{1}{2n} & \text{if } -2 \le x < 0, \\ 1 - \frac{1}{2n} & \text{if } 0 \le x < 2, \\ 1 & \text{if } x \ge 2. \end{cases}$$



Taking n to infinity we obtain

$$\lim_{n\to\infty} F_{X_n}(x) = \begin{cases} 0 & \text{if } x < 0 \\ 1 & \text{if } x \ge 0, \end{cases}$$

which is the CDF of $X \in \delta(0)$.

In particular, F_{X_n} converges for all continuity points of the CDF of X. Thus $X_n \xrightarrow[n\to\infty]{d} X \in \delta(0)$. //

c.) Since X_k's are centered and independent, we have

$$\mathbb{E}[Y_n^2] = \mathbb{E}\left[\sum_{k=1}^{n} \frac{X_k}{\sqrt{k}} \cdot \sum_{i=1}^{n} \frac{X_i}{\sqrt{i}}\right]$$

$$= \sum_{k=1}^{n} \frac{1}{k} \mathbb{E}[X_k^2].$$

For each k, we compute

$$\mathbb{E}[X_k^2] = (-2)^2 \underbrace{\mathbb{P}(X_k=-2)}_{\frac{1}{2k}} + \underbrace{0^2 \mathbb{P}(X_k=0)}_{=0} + 2^2 \underbrace{\mathbb{P}(X_k=2)}_{\frac{1}{2k}}$$

$$= \frac{4}{2k} + \frac{4}{2k} = \frac{2}{k}.$$

Thus $\mathbb{E}[Y_n^2] = \sum_{k=1}^{n} \frac{1}{k} \cdot \frac{2}{k} = 2 \cdot \sum_{k=1}^{n} \frac{1}{k^2}$.

With the hint, we hence get

$$\lim_{n\to\infty} \mathbb{E}[Y_n^2] = \lim_{n\to\infty} 2\sum_{k=1}^{n} \frac{1}{k^2} = 2\sum_{k=1}^{\infty} \frac{1}{k^2} = \frac{\pi^2}{3}. \quad //$$

2.) We write the joint density of $(\bar{X}, \bar{Y})$ as

$$f_{\bar{X},\bar{Y}}(x,y) = 2xy\, e^{-x-y}\, \mathbb{1}(y>x>0).$$

Next, for $x, y > 0$, we obtain from $u = \frac{x}{x+y}$, $v = x+y$, that

$$x = u \cdot v \quad \text{and} \quad y = (1-u)v, \quad \text{where } 0<u<1 \text{ and } v>0.$$

The transformation has Jacobian

$$J = \det\begin{pmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{pmatrix} = \det\begin{pmatrix} v & u \\ -v & (1-u) \end{pmatrix}$$

$$= v(1-u) + vu = v.$$

By the transformation theorem,

$$f_{U,V}(u,v) = \begin{cases} f_{\bar{X},\bar{Y}}(uv, (1-u)v) \cdot |J|, & \text{if } 0<u<1, v>0, \\ 0, & \text{else.} \end{cases}$$

$$= 2uv(1-u)v e^{-v} \mathbb{1}((1-u)v > uv)\, \mathbb{1}(0<u<1, v>0)$$

If $0<u<1$ and $v>0$, then $(1-u)v > uv$ means $(1-u) > u$, $u>0$, hence also $0<u<1/2$.

Therefore we have

$$f_{\bar{U},\bar{V}}(u,v) = 2u(1-u)v^3 e^{-v} \mathbb{1}(v>0)\mathbb{1}(0<u<1/2).$$

b.) If $\bar{U}$ and $\bar{V}$ are independent, then their joint density must factorize into the marginals of $\bar{U}$ and $\bar{V}$. We write the joint density as

$$f_{\bar{U},\bar{V}}(u,v) = \underbrace{2u(1-u)\mathbb{1}(0<u<1/2)}\cdot \underbrace{v^3 e^{-v}\mathbb{1}(v>0)},$$

so the right side factorizes in functions of u and v. It remains to distribute numerical constants (the factor 2).

Note that

$$\int_0^{1/2} u(1-u)\,du = \frac{1}{2}u^2 - \frac{1}{3}u^3\Big|_{u=0}^{1/2} = \frac{1}{8} - \frac{1}{3\cdot 8} = \frac{3-1}{24} = \frac{1}{12}.$$

Thus $12u(1-u)\mathbb{1}(0<u<1/2)$ is a probability density and we can write

$$f_{\bar{U},\bar{V}}(u,v) = \underbrace{12u(1-u)\mathbb{1}(0<u<1/2)}_{=f_{\bar{U}}(u)}\cdot \underbrace{\frac{1}{6}v^3 e^{-v}\mathbb{1}(v>0)}_{=f_{\bar{V}}(v)}.$$

Answer: $\bar{U}$ and $\bar{V}$ are independent, and

$$f_{\bar{U}}(u) = 12u(1-u)\mathbb{1}(0<u<1/2),$$
$$f_{\bar{V}}(v) = \frac{1}{6}v^3 e^{-v}\mathbb{1}(v>0).$$

3.)

a.) We compute the characteristic function of $\bar{Y}$:

$$\varphi_{\bar{Y}}(t) = \mathbb{E}\left[e^{it\bar{Y}}\right] = \mathbb{E}\left[\mathbb{E}\left[e^{it\bar{Y}} \mid \bar{U}\right]\right]$$

law of total expectation

$$= \frac{1}{2}\mathbb{E}\left[e^{it\bar{Y}} \mid \bar{U}=0\right] + \frac{1}{2}\mathbb{E}\left[e^{it\bar{Y}} \mid \bar{U}=1\right]$$

$\bar{U} \in Be(1/2)$

$$= \frac{1}{2}\mathbb{E}\left[e^{it\bar{X}} \mid \bar{U}=0\right] + \frac{1}{2}\mathbb{E}\left[e^{-it\bar{X}} \mid \bar{U}=1\right]$$

def. of $\bar{Y}$

$$= \frac{1}{2}\mathbb{E}\left[e^{it\bar{X}}\right] + \frac{1}{2}\mathbb{E}\left[e^{-it\bar{X}}\right]$$

$\bar{X} \perp\!\!\!\perp \bar{U}$

$$= \frac{1}{2}e^{-\frac{t^2}{2}} + \frac{1}{2}e^{-\frac{(-t)^2}{2}}$$

$X \in N(0,1)$

$$= e^{-\frac{t^2}{2}}.$$

By the uniqueness theorem, $\bar{Y} \in N(0,1)$.

b.) $\bar{X}, \bar{Y} \in N(0,1)$, so $Var(\bar{X}) = Var(\bar{Y}) = 1$. Moreover

$$Cov(\bar{X}, \bar{Y}) = \mathbb{E}[\bar{X}\cdot\bar{Y}]$$

law of total expectation

$$= \mathbb{E}\left[\mathbb{E}[\bar{X}\cdot\bar{Y} \mid \bar{U}]\right]$$

$\bar{U} \in Be(\frac{1}{2})$

$$= \frac{1}{2}\mathbb{E}[\bar{X}\cdot\bar{Y} \mid \bar{U}=0] + \frac{1}{2}\mathbb{E}[\bar{X}\cdot\bar{Y} \mid U=1]$$

def $\bar{Y}$

$$= \frac{1}{2}\mathbb{E}[\bar{X}^2 \mid \bar{U}=0] - \frac{1}{2}\mathbb{E}[\bar{X}^2 \mid U=1]$$

$X \perp\!\!\!\perp U$

$$= \frac{1}{2}\mathbb{E}[\bar{X}^2] - \frac{1}{2}\mathbb{E}[X^2]$$

$$= \frac{1}{2} - \frac{1}{2} = 0.$$

Thus the covariance matrix Λ is given by $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$.

c.) $(\bar{X}, \bar{Y})^t$ is not a Gaussian vector. We give several arguments.

Cramér-Wold: If $(\bar{X}, \bar{Y})$ were Gaussian, then so were any linear combination. But

$$\bar{X} + \bar{Y} = \begin{cases} 2\bar{X}, & \text{if } U = 0 \\ 0, & \text{if } U = 1 \end{cases}$$

is not Gaussian (it is zero with probability one half) //

Characteristic function: We compute

$$\varphi_{\bar{X},\bar{Y}}(s,t) = \mathbb{E}\left[e^{is\bar{X}+it\bar{Y}}\right]$$

i.t.e.

$$= \mathbb{E}\left[\mathbb{E}\left[e^{is\bar{X}+it\bar{Y}} \mid U\right]\right]$$

$U \in Be(\frac{1}{2})$

$$= \frac{1}{2}\mathbb{E}\left[e^{i(s+t)\bar{X}} \mid U=0\right] + \frac{1}{2}\mathbb{E}\left[e^{i(s-t)\bar{X}} \mid U=1\right]$$

$U \perp\!\!\!\perp \bar{X}$

$$= \frac{1}{2}\mathbb{E}\left[e^{i(s+t)\bar{X}}\right] + \frac{1}{2}\mathbb{E}\left[e^{i(s-t)\bar{X}}\right]$$

$$= \frac{1}{2}e^{-\frac{(s+t)^2}{2}} + \frac{1}{2}e^{-\frac{(s-t)^2}{2}}.$$

But this is not the characteristic function of a Gaussian. Choose for example $t=s$, then

$$\varphi_{\bar{X},\bar{Y}}(t,t) = \frac{1}{2}e^{-2t^2} + \frac{1}{2}. \quad //$$

Characteristic function 2: In part 6.) we found that the covariance matrix is $\Lambda = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, hence if $(\bar{X},\bar{Y})'$ were Gaussian, we had

$$\varphi_{\bar{X},\bar{Y}}(s,t) = e^{-\frac{1}{2}t^2 - \frac{1}{2}s^2},$$

which is not what we computed above. //

Independence: In part 5.) we found that Λ is diagonal, hence if $(\bar{X},\bar{Y})$ were Gaussian, $\bar{X}$ and $\bar{Y}$ would be independent, but $\bar{Y}$ is a non-constant function of $\bar{X}$. //

Remark:
- We found an example of bivariate random vector, whose marginal distributions are Gaussian, but the joint distribution is not.
- $\bar{X}$ and $\bar{Y}$ are uncorrelated, but not independent.

4.) a.) We have $N_n \in Ge(p_n)$. From the table it has characteristic function

$$\varphi_{N_n}(t) = \frac{p_n}{1-(1-p_n)e^{it}}.$$

Thus $p_n N_n$ has characteristic function

$$\varphi_{p_n N_n}(t) = \varphi_{N_n}(p_n t) = \frac{p_n}{1-(1-p_n)e^{itp_n}}.$$

Now as $n \to \infty$, we have $p_n \to 0$, thus

$$e^{itp_n} = 1 + itp_n + o(p_n).$$

Thus, as $n \to \infty$, we have

$$\varphi_{p_n N_n}(t) = \frac{p_n}{1-(1-p_n)(1+itp_n+o(p_n))}$$

$$= \frac{p_n}{1-(1+itp_n-p_n+o(p_n))}$$

$$= \frac{p_n}{p_n - itp_n + o(p_n)}$$

$$= \frac{1}{1-it+o(1)}.$$

Thus as $n \to \infty$,

$$\lim_{n\to\infty} \varphi_{p_n N_n}(t) = \frac{1}{1-it}.$$

By uniqueness theorem, $p_n N_n$ converges in distribution to an $Exp(1)$ random variable, as $n \to \infty$.

b.) We first note that

$$\varphi_{p_n S_{N_n}}(t) = \varphi_{S_{N_n}}(p_n t).$$

Since X_j's are i.i.d. and independent from N_n, we have the composition formula

$$\varphi_{S_{N_n}}(p_n t) = g_{N_n}\left(\varphi_{X_1}(p_n t)\right),$$

where $g_{N_n}(\cdot)$ is the probability generating function of N_n, i.e.

$$\begin{aligned} g_{N_n}(s) &= \sum_{k=0}^{\infty} s^k \, \mathbb{P}(N_n = k) \\ &= \sum_{k=0}^{\infty} s^k p_n (1-p_n)^k \\ &= p_n \sum_{k=0}^{\infty} \left(s(1-p_n)\right)^k \\ &= \frac{p_n}{1 - s(1-p_n)}, \quad \text{for } |s| \le 1. \end{aligned}$$

Thus $\varphi_{S_{N_n}}(p_n t) = g_{N_n}\left(\varphi_{X_1}(p_n t)\right) = \dfrac{p_n}{1 - \varphi_{X_1}(p_n t)(1-p_n)}$, since $|\varphi_{X_1}(\cdot)| \le 1$.

As $p_n \to 0$, we have

$$\begin{aligned} \varphi_{X_1}(p_n t) &= 1 + i t p_n \underbrace{\mathbb{E}[X_1]}_{=\mu} + o(p_n) \\ &= 1 + i t p_n \mu + o(p_n). \end{aligned}$$

Therefore, as $p_n \to 0$,

$$\varphi_{S_{N_n}}(p_n t) = \frac{p_n}{1 - (1 + it p_n \mu + o(p_n))(1 - p_n)}$$

$$= \frac{p_n}{1 - (1 + it p_n \mu - p_n + o(p_n))}$$

$$= \frac{p_n}{p_n - it p_n \mu + o(p_n)}.$$

As $p_n \to 0$ we therefore get

$$\lim_{n \to \infty} \varphi_{S_{N_n}}(p_n t) = \frac{1}{1 - it\mu}.$$

By weak limit theorem, $p_n S_{N_n}$ converges to an Exponential random variable with parameter $\mu = \mathbb{E}[X_1] > 0$.

//